Spring Semester 2025–2026

MATH3026/4022: Time Series Analysis/Time Series and Forecasting
Exercises Sheet 1 - SOLUTIONS

1. Suppose that X_t is a variable from a time series at time t. Apply the following operators to X_t:

   (a) $B - B^3 + B^5$

   SOLUTION:

   $$(B - B^3 + B^5)X_t = X_{t-1} - X_{t-3} + X_{t-5}$$

   (b) $\psi(B) = 1 - \frac{1}{2}B + \frac{1}{4}B^2 - \frac{1}{6}B^4$

   SOLUTION:

   $$\psi(B)X_t = X_t - \frac{1}{2}X_{t-1} + \frac{1}{4}X_{t-2} - \frac{1}{6}X_{t-4}$$

   (c) $\nabla(1 + B - B^2)$

   SOLUTION:

   $$\begin{aligned}\nabla(1 + B - B^2)X_t &= \nabla[X_t + X_{t-1} - X_{t-2}] \\ &= (X_t + X_{t-1} - X_{t-2}) - (X_{t-1} + X_{t-2} - X_{t-3}) \\ &= X_t - 2X_{t-2} + X_{t-3}.\end{aligned}$$

2. Write the following process in $MA(\infty)$ form

   $$\left(1 - \frac{1}{3}B\right) X_t = Z_t.$$

   SOLUTION:

   $$\begin{aligned}\left(1 - \frac{1}{3}B\right) X_t &= Z_t \\ \implies X_t &= \frac{1}{1 - \frac{1}{3}B} Z_t\end{aligned}$$

Let $f(B) = \left(1 - \frac{1}{3}B\right)^{-1}$, then

$$\begin{aligned} f'(B) &= \frac{1}{3}\left(1 - \frac{1}{3}B\right)^{-2} \\ f''(B) &= \left(\frac{1}{3}\right)^2 2! \left(1 - \frac{1}{3}B\right)^{-3} \\ f'''(B) &= \left(\frac{1}{3}\right)^3 3! \left(1 - \frac{1}{3}B\right)^{-4} \end{aligned}$$

Generally, the k^{th} derivative of $f(B)$ is

$$f^{(k)}(B) = \left(\frac{1}{3}\right)^k k! \left(1 - \frac{1}{3}\right)^{-(k+1)}.$$

The Taylor expansion of $f(B)$ (about 0) is

$$\begin{aligned} f(B) &= f(0) + f'(0)B + \frac{1}{2!}f''(0)B^2 + \frac{1}{3!}f'''(0)B^3 + \dots \\ &= 1 + \frac{1}{3}B + \left(\frac{1}{3}\right)^2 B^2 + \left(\frac{1}{3}\right)^3 B^3 + \dots \\ &= \sum_{j=0}^{\infty} \left(\frac{1}{3}\right)^j B^j \end{aligned}$$

It follows that

$$\begin{aligned} X_t &= \frac{1}{1 - \frac{1}{3}B} Z_t \\ &= \sum_{j=0}^{\infty} \left(\frac{1}{3}\right)^j B^j Z_t \\ &= \sum_{j=0}^{\infty} \left(\frac{1}{3}\right)^j Z_{t-j}. \end{aligned}$$

3. A mean zero stationary process $\{X_t\}$ has autocovariance function $\gamma_X(h)$. Suppose that $\{Y_t\}$ is a new process, defined

$$Y_t = X_t - X_{t-1}.$$

(a) Show that the process $\{Y_t\}$ has mean zero for all t.

SOLUTION:

Clearly

$$\begin{aligned} \mathbb{E}(Y_t) &= \mathbb{E}(X_t - X_{t-1}) \\ &= \mathbb{E}(X_t) - \mathbb{E}(X_{t-1}) \\ &= 0 \quad \text{(since } X_t \text{ has mean zero for all } t\text{)} \end{aligned}$$

(b) Determine the autocovariance function of $\{Y_t\}$ in terms of $\gamma_X(h)$.

SOLUTION:

For $h = 0, \pm 1, \pm 2, \ldots$

$$\begin{aligned}
\gamma_Y(h) &= \text{Cov}(Y_t, Y_{t+h}) \\
&= \mathbb{E}\left[(Y_t - \mathbb{E}(Y_t))(Y_{t+h} - \mathbb{E}(Y_{t+h}))\right] \\
&= \mathbb{E}(Y_t Y_{t+h}) \quad (\text{since } \mathbb{E}(Y_t) = 0 \text{ for all } t) \\
&= \mathbb{E}\left[(X_t - X_{t-1})(X_{t+h} - X_{t+h-1})\right] \\
&= \mathbb{E}(X_t X_{t+h}) - \mathbb{E}(X_{t-1} X_{t+h}) - \mathbb{E}(X_t X_{t+h-1}) + \mathbb{E}(X_{t-1} X_{t+h-1}) \\
&= \gamma_X(h) - \gamma_X(h+1) - \gamma_X(h-1) + \gamma_X(h) \\
&= 2\gamma_X(h) - \gamma_X(h+1) - \gamma_X(h-1).
\end{aligned}$$

4. Suppose that $\{Z_t\}$ is a white noise process with variance σ_z^2.

(a) Show that the $MA(\infty)$ process, $\{X_t\}$, defined

$$X_t = Z_t + C(Z_{t-1} + Z_{t-2} + \ldots)$$

where C is constant, is non-stationary where $C \neq 0$.

SOLUTION:

Given that $X_t = Z_t + C(Z_{t-1} + Z_{t-2} + \ldots)$, we recognise this as an $MA(\infty)$ process with the form

$$X_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}$$

where $\psi_0 = 1$ and $\psi_j = C$ for $j \geq 1$.

From lectures, we know that a necessary condition for the above process to be stationary is that

$$\sum_{j=0}^{\infty} \psi_j^2 < \infty.$$

However,

$$\sum_{j=0}^{\infty} \psi_j^2 = 1 + C^2 + C^2 + C^2 + \ldots$$

is not finite unless $C = 0$. Hence, the time series is not stationary unless $C = 0$.

If $C = 0$ then the process is white noise which is clearly stationary.

(b) Show that the process $\{Y_t\}$, defined

$$Y_t = X_t - X_{t-1}$$

is a stationary moving average process and hence determine the autocorrelation function of $\{Y_t\}$.

SOLUTION:

Now, suppose that we define $Y_t = X_t - X_{t-1}$. Then

$$\begin{aligned}
Y_t &= X_t - X_{t-1} \\
&= Z_t + C(Z_{t-1} + Z_{t-2} + \ldots) - [Z_{t-1} + C(Z_{t-2} + Z_{t-3} + \ldots)] \\
&= Z_t + (C-1)Z_{t-1}.
\end{aligned}$$

We see that $\{Y_t\}$ is an $MA(1)$ process with $\theta_1 = C - 1$. For the autocovariance function:

$$\begin{aligned}
\gamma(0) &= \mathrm{Var}(Y_t) \\
&= \mathrm{Var}(Z_t + (C-1)Z_{t-1}) \\
&= \mathrm{Var}(Z_t) + (C-1)^2\mathrm{Var}(Z_{t-1}) + 2(C-1)\mathrm{Cov}(Z_t, Z_{t-1}) \\
&= \sigma_Z^2 \left(1 + (C-1)^2\right).
\end{aligned}$$

At lag 1

$$\begin{aligned}
\gamma(1) &= \mathrm{Cov}(Y_t, Y_{t-1}) \\
&= \mathbb{E}(Y_t Y_{t-1}) \quad (\text{since } \mathbb{E}(Y_t) = 0 \ \forall t) \\
&= \mathbb{E}[(Z_t + (C-1)Z_{t-1})(Z_{t-1} + (C-1)Z_{t-2})] \\
&= \mathbb{E}(Z_t Z_{t-1}) + (C-1)\left[\mathbb{E}(Z_{t-1}^2) + \mathbb{E}(Z_t Z_{t-2})\right] + (C-1)^2\mathbb{E}(Z_t Z_{t-2}) \\
&= \sigma_Z^2(C-1).
\end{aligned}$$

Since Y_t is an $MA(1)$ process, $\gamma(h) = 0$ for $|h| > 1$. Hence

$$\gamma(h) = \begin{cases} \sigma_Z^2\left(1 + (C-1)^2\right) & h = 0; \\ \sigma_Z^2(C-1) & h = \pm 1; \\ 0 & |h| > 1. \end{cases}$$

The autocorrelation function is

$$\gamma(h) = \begin{cases} 1 & h = 0; \\ \frac{C-1}{1+(C-1)^2} & h = \pm 1; \\ 0 & |h| > 1. \end{cases}$$

5. The process $\{Z_t\}$ is a white noise process with variance σ_z^2 and the process $\{X_t\}$ is defined

$$X_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}.$$

(a) Show that $\mathbb{E}(X_t Z_{t-h}) = \sigma_z^2 \psi_h$.

SOLUTION:

Multiplying X_t by Z_{t-h} and taking expectations, we obtain

$$\mathbb{E}(X_t Z_{t-h}) = \sum_{j=0}^{\infty} \psi_j \mathbb{E}(Z_{t-j} Z_{t-h}).$$

We know that $\mathbb{E}(Z_{t-j} Z_{t-h}) = 0$ where $j \neq h$ because $\{Z_t\}$ is a white noise process. Hence

$$\begin{aligned} \mathbb{E}(X_t Z_{t-h}) &= \psi_h \mathbb{E}(Z_{t-h} Z_{t-h}) \\ &= \psi_h \sigma_z^2 \quad \text{as required.} \end{aligned}$$

(b) Show that $\mathbb{E}(X_t X_{t-h}) = \sigma_z^2 \sum_{t=h}^{\infty} \psi_j \psi_{j-h}$.

SOLUTION:

$$\begin{aligned} \mathbb{E}(X_t X_{t-h}) &= \mathbb{E}\left(\left[\sum_{j=0}^{\infty} \psi_j Z_{t-j} \right] \left[\sum_{k=0}^{\infty} \psi_k Z_{t-h-k} \right] \right) \\ &= \left(\sum_{j=0}^{\infty} \sum_{k=0}^{\infty} \psi_j \psi_k \mathbb{E}(Z_{t-j} Z_{t-h-k}) \right). \end{aligned}$$

Non-zero terms in the above expectation will occur where

$$\begin{aligned} t - j &= t - h - k \\ \implies j &= h + k \\ \implies k &= j - h. \end{aligned}$$

Since $k \geq 0$, it follows that $j \geq h$. Therefore

$$\begin{aligned} \mathbb{E}(X_t X_{t-h}) &= \sum_{j=h}^{\infty} \psi_j \psi_{j-h} \mathbb{E}(Z_{t-j}^2) \\ &= \sigma_z^2 \sum_{j=h}^{\infty} \psi_j \psi_{j-h}. \end{aligned}$$

6. Consider the following $MA(1)$ processes:

$$X_t = Z_t + \theta Z_{t-1}$$
$$Y_t = W_t + \frac{1}{\theta} W_{t-1}$$

where $\{Z_t\}$ and $\{W_t\}$ are mean zero white noise processes with variances σ^2 and $\theta^2\sigma^2$, respectively, with $|\theta| < 1$.

(a) Show that $\{X_t\}$ and $\{Y_t\}$ have the same autocovariance function.

SOLUTION:

For X_t

$$\begin{aligned} \mathrm{Var}(X_t) &= \mathrm{Var}(Z_t + \theta Z_{t-1}) \\ &= \mathrm{Var}(Z_t) + \theta^2 \mathrm{Var}(\theta Z_{t-1}) \\ &= \sigma^2(1+\theta^2). \end{aligned}$$

Also

$$\begin{aligned} \mathrm{Cov}(X_t, X_{t-1}) &= \mathbb{E}\left[(Z_t + \theta Z_{t-1})(Z_{t-1} + \theta Z_{t-2})\right] \\ &= \theta \mathbb{E}(Z_{t-1}^2) \\ &= \theta\sigma^2. \end{aligned}$$

Similarly

$$\begin{aligned} \mathrm{Var}(Y_t) &= \mathrm{Var}(W_t + \frac{1}{\theta}\theta W_{t-1}) \\ &= \mathrm{Var}(W_t) + \frac{1}{\theta^2}\mathrm{Var}(W_{t-1}) \\ &= \theta^2\sigma^2 + \frac{1}{\theta^2}\theta^2\sigma^2 \\ &= \sigma^2(1+\theta^2). \end{aligned}$$

Also

$$\begin{aligned} \mathrm{Cov}(Y_t, Y_{t-1}) &= \mathbb{E}\left[\left(W_t + \frac{1}{\theta}W_{t-1}\right)\left(W_{t-1} + \frac{1}{\theta}W_{t-2}\right)\right] \\ &= \frac{1}{\theta}\mathbb{E}(W_{t-1}^2) \\ &= \frac{1}{\theta}\theta^2\sigma^2 \\ &= \theta\sigma^2. \end{aligned}$$

We see that the autocovariance function is the same for the processes $\{X_t\}$ and $\{Y_t\}$ with

$$\gamma_X(h) = \gamma_Y(h) = \begin{cases} \sigma^2(1+\theta^2) & h = 0; \\ \theta\sigma^2 & h = \pm 1; \\ 0 & |h| \geq 2. \end{cases}$$

(b) Determine the behaviour of $\{X_t\}$ and $\{Y_t\}$ for

(i) $\theta \to 0$

SOLUTION:

As $\theta \to 0$ the process $\{X_t\}$ is such that

$$X_t = Z_t + \theta Z_{t-1} \to Z_t.$$

We can show this using convergence in mean square.

$$\begin{aligned}
\mathbb{E}[(X_t - Z_t)^2] &= \mathbb{E}[Z_t + \theta Z_{t-1} - Z_t)^2] \\
&= \theta^2 \mathbb{E}(Z_{t-1}^2) \\
&= \theta^2 \sigma^2
\end{aligned}$$

and

$$\lim_{\theta \to 0} \mathbb{E}[(X_t - Z_t)^2] = 0.$$

Also, Y_t converges to a white noise process. Let $V_t = W_t/\theta$ which is a white noise process with variance σ^2. Then

$$\begin{aligned}
\mathbb{E}[(Y_t - V_{t-1})^2] &= \mathbb{E}\left[\left(W_t + \frac{W_{t-1}}{\theta} - V_{t-1}\right)^2\right] \\
&= \mathbb{E}\left[(\theta V_t + V_{t-1} - V_{t-1})^2\right] \\
&= \theta^2 \mathbb{E}(V_t^2) \\
&= \theta^2 \mathbb{E}\left(\frac{W_t^2}{\theta^2}\right) \\
&= \mathbb{E}(W_t^2) \\
&= \theta^2 \sigma^2 \to 0 \quad \text{as } \theta \to 0.
\end{aligned}$$

(ii) $\theta = 1$

SOLUTION:

If $\theta = 1$ then both processes are still $MA(1)$ processes but neither is invertible.

7. The process $\{X_t\}$ is defined

$$X_t = U \cos(\omega t) + V \sin(\omega t) \quad t = 0, \pm 1, \pm 2, \ldots$$

where U and V are independent random variables, each with mean zero and variance 1, and $\omega \in [0, \pi]$ is a constant.

(a) Show that the process $\{X_t\}$ is stationary and determine the mean, autocovariance and autocorrelation function for this process.

SOLUTION:

$$\begin{aligned}
\mathbb{E}(X_t) &= \mathbb{E}\left[U\cos(\omega t) + V\sin(\omega t)\right] \\
&= \cos(\omega t)\mathbb{E}(U) + \sin(\omega t)\mathbb{E}(V) \\
&= 0 \ \text{ because } \mathbb{E}(U) = \mathbb{E}(V) = 0.
\end{aligned}$$

Now, we consider the autocovariance to show that process is stationary.

$$\begin{aligned}
\mathrm{Cov}(X_t, X_{t+h}) &= \mathbb{E}\left[(U\cos(\omega t) + V\sin(\omega t))\left(U\cos(\omega(t+h)) + V\sin(\omega(t+h))\right)\right] \\
&= \mathbb{E}(U^2)\cos(\omega t)\cos(\omega(t+h)) + \mathbb{E}(V^2)\sin(\omega t)\sin(\omega(t+h)) \\
&\quad + \mathbb{E}(UV)\left[\cos(\omega t)\sin(\omega(t+h)) + \sin(\omega t)\cos(\omega(t+h))\right] \\
&= \cos(\omega t)\cos(\omega(t+h)) + \sin(\omega t)\sin(\omega(t+h))
\end{aligned}$$

We note that

$$\begin{aligned}
\sin(\omega(t+h)) &= \sin(\omega t)\cos(\omega h) + \cos(\omega t)\sin(\omega h); \\
\cos(\omega(t+h)) &= \cos(\omega t)\cos(\omega h) - \sin(\omega t)\sin(\omega h).
\end{aligned}$$

Hence

$$\begin{aligned}
\gamma(h) = \mathrm{Cov}(X_t, X_{t+h}) &= \cos(\omega t)\left[\cos(\omega t)\cos(\omega h) - \sin(\omega t)\sin(\omega h)\right] \\
&\quad + \sin(\omega t)\left[\sin(\omega t)\cos(\omega h) + \cos(\omega t)\sin(\omega h)\right] \\
&= \cos(\omega h)\left[\sin^2(\omega t) + \cos^2(\omega t)\right] \\
&= \cos(\omega h).
\end{aligned}$$

We see that the autocovariance function depends only on the lag, h, and not on t so the process is stationary. The autocorrelation function is

$$\begin{aligned}
\rho(h) &= \frac{\gamma(h)}{\gamma(0)} \\
&= \frac{\cos(\omega h)}{1} \\
&= \cos(\omega h) \quad h = 0, \pm 1, \pm 2, \ldots
\end{aligned}$$

(b) Suppose now that U and V are now correlated where $\mathrm{Cov}(U, V) = \xi \neq 0$ but that each of U and V still has mean zero and variance 1. Is the process $\{X_t\}$ still stationary? Justify your answer.

SOLUTION:

Suppose that $\mathrm{Cov}(U, V) = \xi$ then consider the variance of X_t

$$\begin{aligned}\mathrm{Var}(X_t) &= \mathrm{Var}(U\cos(\omega t) + V\sin(\omega t)) \\ &= \cos^2(\omega t)\mathrm{Var}(U) + \sin^2(\omega t)\mathrm{Var}(V) + 2\sin(\omega t)\cos(\omega t)\mathrm{Cov}(U, V) \\ &= \cos^2(\omega t) + \sin^2(\omega t) + 2\xi\sin(\omega t)\cos(\omega t) \\ &= 1 + 2\xi\sin(\omega t)\cos(\omega t) \\ &= 1 + \xi\sin(2\omega t).\end{aligned}$$

Clearly, $\mathrm{Var}(X_t)$ depends on t when $\xi \neq 0$ which implies that the process is not stationary when $\xi \neq 0$.

8. Consider the process

$$X_t = 5 + 3t + Z_t - \frac{1}{2}Z_{t-1}$$

where $\{Z_t\}$ is a white noise process with variance σ_z^2.

(a) Is this process stationary? Justify your answer.

SOLUTION:

We see that

$$\begin{aligned}\mathbb{E}(X_t) &= 5 + 3t + \mathbb{E}(Z_t) - \frac{1}{2}\mathbb{E}(Z_{t-1}) \\ &= 5 + 3t.\end{aligned}$$

So the mean of the process is not constant and, hence, the process is not stationary.

(b) Show that the differenced process $Y_t = X_t - X_{t-1}$ is an $MA(2)$ process where

$$Y_t = \mu + Z_t + a_1 Z_{t-1} + a_2 Z_{t-2}$$

and state the values of μ, a_1 and a_2.

SOLUTION:

Let

$$\begin{aligned}Y_t &= X_t - X_{t-1} \\ &= \left(5 + 3t + Z_t - \frac{1}{2}Z_{t-1}\right) - \left(5 + 3(t-1) + Z_{t-1} - \frac{1}{2}Z_{t-2}\right) \\ &= 3 + Z_t - \frac{3}{2}Z_{t-1} + \frac{1}{2}Z_{t-2}\end{aligned}$$

which is an $MA(2)$ process with $\mu = 3$, $a_1 = -\frac{3}{2}$ and $a_2 = \frac{1}{2}$.

(c) Suppose that $a_2 = \frac{1}{2}$ but a_1 is unknown. Determine the range of values of a_1 for which the process $\{Y_t\}$ is invertible.

SOLUTION:

We see that

$$\begin{aligned} Y_t - \mu &= Z_t + a_1 Z_{t-2} + a_2 Z_{t-2} \\ &= \left(1 + a_1 B + a_2 B^2\right) Z_t \end{aligned}$$

and the process $\{Y_t\}$ is invertible if the roots of

$$1 + a_1 z + a_2 z^2 = 0$$

lie outside the unit circle. The roots are given by

$$\frac{-a_1 \pm \sqrt{a_1^2 - 4a_2}}{2a_2}$$

If $a_2 = \frac{1}{2}$ then we have

$$-a_1 \pm \sqrt{a_1^2 - 2}.$$

Now consider the cases $|a_1| \leq \sqrt{2}$ and $|a_1| > \sqrt{2}$ separately.

If $|a_1| \leq \sqrt{2}$ then

$$\begin{aligned} \left| -a_1 \pm \sqrt{a_1^2 - 2} \right|^2 &= a_1^2 + (2 - a_1^2) \\ &= 2 \end{aligned}$$

so the roots lie outside the unit circle where $|a_1| \leq \sqrt{2}$.

Now suppose $|a_1| > \sqrt{2}$, then, if the roots lie outside the unit circle, it follows that

$$\begin{aligned} |a_1| - \sqrt{a_1^2 - 2} &> 1 \\ \implies |a_1| - 1 &> \sqrt{a_1^2 - 2} \\ \implies |a_1|^2 - 2|a_1| + 1 &> |a_1|^2 - 2 \\ \implies -2|a_1| &> -3 \\ \implies |a_1| &< \frac{3}{2}. \end{aligned}$$

Thus the process is invertible where $|a_1| \leq \sqrt{2}$ and $|a_1| < \frac{3}{2}$. Since $\sqrt{2} < \frac{3}{2}$ then we say that the process is invertible if $|a_1| < \frac{3}{2}$.